

Unit 9, Lesson 1: Writing a Quadratic Equation Given x -Intercepts and a Point



Benchmarks

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

MA.912.AR.3.5 Given the x -intercepts and another point on the graph of a quadratic function, write the equation for the function.



Learning Targets

- ☐ I can rearrange a quadratic equation to isolate a quantity of interest.
- ☐ I can write an equation given the x -intercepts and another point on the graph of a quadratic function.



9.1.1 Warm-Up: Bell Work

1. Solve for C . $F = \frac{9}{5}C + 32$

2. Solve for v . $4(2v - 3w) = 7v + 5w$



9.1.2 Exploration: Using a Point and x -Intercepts to Find “ a ” in a Quadratic Equation

Jaxon’s teacher gave him a card with the following information.

Equation	$y = a(x - 1)(x - 4)$
Point on the Parabola	$(2, -6)$

1. In what form is the quadratic equation on Jaxon’s card?
2. Discuss with your partner the significance of the points $(1, 0)$ and $(4, 0)$ and how they are related to the form of the quadratic. Summarize your discussion.
3. Discuss with your partner if it is possible to know if a given point is on a parabola. Justify your reasoning.
4. Explain why it is not yet possible to verify that $(2, -6)$ is on the parabola.

5. Jaxon’s first step for solving for the variable a is shown.

$$-6 = a(2 - 1)(2 - 4)$$

Explain what Jaxon did in his first step.

6. Use $-6 = a(2 - 1)(2 - 4)$ to find the value of a .

7. Work with your partner to write the equation of the quadratic that contains the points $(1, 0)$, $(4, 0)$, and $(2, -6)$ in both factored and standard form.

Factored Form	Standard Form

8. Discuss with your partner which form is easier to use when finding other points on the parabola. Specifically address whether or not it matters if the equation is written in standard form or factored form.

9. With another classmate, compare your answers and equations. Make adjustments if needed.

10. Use your equation to determine if the points $(0, 12)$ and $(-1, -30)$ on the parabola.

11. Describe how to determine if points are on the parabola.



9.1.3 Guided Instruction: Writing Quadratic Equations From Intercepts and a Point

A parabola has the x -intercepts $(9, 0)$ and $(3, 0)$. The point $(4, 10)$ also lies on the parabola.

When given the x -intercepts and another point on the parabola, it is possible to write the equation of the quadratic. Because the x -intercepts are given, the factored form of a quadratic is used to write the equation of the function.

Step 1: Substitute in the values provided for the x -intercepts.	$y = a(x - p)(x - q)$ $y = a(x - 9)(x - 3)$
Step 2: Substitute in the values provided for the point (x, y) .	$10 = a(4 - 9)(4 - 3)$
<i>Note: The order of the first two steps can be reversed.</i>	
Step 3: Solve for the variable a .	$10 = a(-5)(1)$ $10 = -5a$ $-2 = a$
Step 4: Use the a -value to write the quadratic function that has x -intercepts $(9, 0)$ and $(3, 0)$ and includes point $(4, 10)$.	$y = -2(x - 9)(x - 3)$

The equation $y = -2(x - 9)(x - 3)$ can be rewritten into standard form.

$$y = -2x^2 + 24x - 54$$

1. Use the equation to find one other point on the parabola.

2. Compare the point you found with your partner. If you found different points, verify both of them.



9.1.4 Guided Practice: Writing Quadratic Equations

1. Sasha is writing the equation of a quadratic function that has x -intercepts $(4, 0)$ and $(-2, 0)$ and passes through the point $(5, -7)$. Her steps are listed below.

Step 1	$-7 = a(5 + 4)(5 - 2)$
Step 2	$-7 = a(9)(3)$ $-7 = 27a$ $a = -\frac{7}{27}$
Step 3	$y = -\frac{7}{27}(x - 4)(x + 2)$

- a. Sasha made an error in her process. With your partner, discuss what error she made.
- b. Revise Sasha's work and write the correct quadratic equation in factored form.

2. A quadratic function has the x -intercepts $(5, 0)$ and $(-5, 0)$. The point $(-4, -36)$ lies on the parabola.

a. Find the value of a .

b. Write the equation of quadratic function in factored form.

c. Rewrite the equation into standard form.

d. Write the equation for a quadratic function that has x -intercepts $(5, 0)$ and $(-5, 0)$ and passes through the point $(6, 44)$.

e. Discuss with your partner what is observed about the equation that passes through $(-4, -36)$ and the equation that passes through $(6, 44)$, where both equations have $(5, 0)$ and $(-5, 0)$ x -intercepts. Summarize your discussion.

3. Write an equation in factored form for the quadratic function with x -intercepts $(-\frac{1}{4}, 0)$ and $(8, 0)$ that passes through point $(0, 2)$.





9.1.5 Your Turn!

1. Show how to find the value of a and write each quadratic in factored form given the x -intercepts and a point.

x -intercepts	Point	Steps to Find " a "	Quadratic Equation in Factored Form
$(7, 0)$ and $(-8, 0)$	$(8, -16)$		
$(4, 0)$ and $(4, 0)$	$(-2, 12)$		
$(\frac{2}{3}, 0)$ and $(-3, 0)$	$(3, 14)$		



9.1.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on writing quadratic equations?

